

Mathematical Models & Decision Errors

IMS Chapters 13 & 14

Recap: from simulation to formula

Where we've been

- ▶ **Ch. 11 (randomization):** simulate the null by shuffling/permuting.
- ▶ **Ch. 12 (bootstrapping):** simulate variability by resampling with replacement.
- ▶ **Today (Ch. 13):** the *same* conclusions, but using a mathematical formula (the normal distribution) instead of running thousands of simulations.

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This works because of something you've already seen: the **Central Limit Theorem**. Sample proportions (and many other statistics) tend to follow a normal distribution when the sample is big enough and observations are independent.

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- ▶ **Standard error (SE)**: the standard deviation of a *statistic's* sampling distribution (not of the raw data).
- ▶ **Margin of error**: $z^* \times SE$ — how far, in a given direction, we expect a point estimate to land from the truth at a chosen confidence level. For 95% confidence, $z^* = 1.96$.

Z-score for a hypothesis test

i Test statistic (Z)

$$Z = \frac{\text{point estimate} - \text{null value}}{SE}$$

This is the exact same Z-score idea from individual observations — just applied to a *statistic* instead of a single data point.

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Once we have Z , we find the p-value the same way we always have: the tail area of $N(0, 1)$ beyond Z .

Case study: opportunity cost, revisited

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Compare to the randomization p-value from Ch. 11: ≈ 0.006 . **Close agreement.**

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Compare to the bootstrap-based simulation p-value: ≈ 0.122 . **Noticeably different this time** — the null distribution here is visibly skewed, so the normal curve is a poor fit. Lesson: **always check conditions** before trusting the normal model.

Building a confidence interval

i 95% confidence interval

$$\text{point estimate} \pm Z_{\alpha} \times SE$$

Z_{α} comes directly from the standard normal distribution: the area between $Z_{2.5} = -1.96$ and $Z_{97.5} = 1.96$ is exactly 0.95.

Case study: the stent study

An experiment: does a brain stent reduce stroke risk after CPR? $n = 451$ patients.

	No event	Stroke	Total
Control	214	13	227
Treatment	191	33	224

Point estimate: $\hat{p}_{trmt} - \hat{p}_{ctrl} = 0.090$ (surprising — stents look *worse*). Given $SE = 0.028$:

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Since this interval is entirely above 0, and this was a randomized **experiment**, we have convincing evidence the stent **caused** increased stroke risk.

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Out of every 100 properly constructed 95% confidence intervals, we expect about 5 to miss the true parameter — not due to bad science, just natural sample-to-sample variability. A confidence interval only tries to capture the **population parameter** — it says nothing about individual observations.

Practice: Z-scores

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$$Z_{Verbal} = \frac{160 - 151}{7} = 1.29 \quad Z_{Quant} = \frac{157 - 153}{7.67} = 0.52$$

. . . .

Her Verbal score is further above its mean (in SD units) — **relatively stronger on Verbal**, even though her raw Quant-adjacent score might look closer to “average” on a different scale.

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Careful: does this interval provide discernible evidence that a *minority* of US adults have a chronic condition (i.e., that the true proportion is below 50%)? Since the entire interval is below 0.50, yes.

Decision errors

Hypothesis tests can be wrong

Just like a court can convict an innocent person or free a guilty one, a hypothesis test can reach the wrong conclusion.

Truth ↓ / Test says →	Reject H_0	Fail to reject H_0
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- ▶ **Type I error:** rejecting H_0 when it's actually true (a “false positive”).
- ▶ **Type II error:** failing to reject H_0 when H_A is actually true (a “false negative”).

Errors in context

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How would we lower the Type I error rate in a court system? What happens to the Type II error rate?

Raise the standard of evidence (e.g., “beyond a reasonable doubt” → “beyond a conceivable doubt”). Fewer innocent people convicted (lower Type I) — but also harder to convict the truly guilty (**higher** Type II). **The two error rates trade off against each other.**

The same idea, medical-testing language

Diagnostic testing uses its own vocabulary for these same two error rates — you'll see this constantly outside the classroom (medicine, spam filters, fraud detection).

	Condition present (H_A true)	Condition absent (H_0 true)
Test positive (reject H_0)	True positive — rate = sensitivity	False positive — rate = α (Type I)
Test negative (fail to reject H_0)	False negative — rate = β (Type II)	True negative — rate = specificity

Testing Language

i Sensitivity and specificity

Sensitivity = $P(\text{test positive} \mid \text{condition present})$ = true positive rate = $1 - \beta$ = power.

Specificity = $P(\text{test negative} \mid \text{condition absent})$ = true negative rate = $1 - \alpha$.

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A “highly sensitive” test rarely misses true cases (low Type II/false-negative rate). A “highly specific” test rarely flags healthy people (low Type I/false-positive rate) — exactly the same tradeoff as choosing α .

Why high accuracy can still mislead

Recall **Bayes' theorem** from our probability review

A disease affects 1% of a population. A screening test has 90% sensitivity and 95% specificity. If someone tests positive, what's the probability they actually have the disease?

$$\begin{aligned} P(\text{disease} \mid +) &= \frac{P(+ \mid \text{disease}) P(\text{disease})}{P(+ \mid \text{disease}) P(\text{disease}) + P(+ \mid \text{no disease}) P(\text{no disease})} \\ &= \frac{(0.90)(0.01)}{(0.90)(0.01) + (0.05)(0.99)} = \frac{0.009}{0.0585} \approx 0.154 \end{aligned}$$

Interpreting the result

Even with a “90% accurate, 95% accurate” test, only about **15.4%** of people who test positive actually have the disease.

- ▶ This isn't a flaw in the test — it's a consequence of the disease being **rare**. With so few true cases, even a small false-positive rate ($\alpha = 0.05$) among the huge healthy majority produces more false positives in absolute terms than true positives among the small diseased group.

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- ▶ This is the same lesson as the discernibility-level discussion: the practical meaning of a positive result depends not just on the test's error rates, but on the **context** — here, how common the condition is to begin with.
- ▶ This is also exactly why we conditioned on the *right* thing back in Ch. 12's medical consultant example — direction of conditioning matters, and it's easy to accidentally answer $P(+ \mid \text{disease})$ when the question actually asked for $P(\text{disease} \mid +)$.

Choosing a discernibility level (α)

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- ▶ If a **Type II error is more dangerous** (e.g., missing a real safety issue), use a *larger* α (0.10) — don't require as much evidence.

One-sided vs. two-sided hypotheses

- ▶ **One-sided test:** H_A specifies a direction (e.g., $p_T - p_C > 0$). Appropriate only when we truly care about just one direction, decided *before* seeing the data.
- ▶ **Two-sided test:** H_A allows *either* direction ($p_T - p_C \neq 0$). This is the **default**, safer choice — keep an open mind.

i Default to two-sided

Use a one-sided test only if you have genuine, pre-specified interest in just one direction.

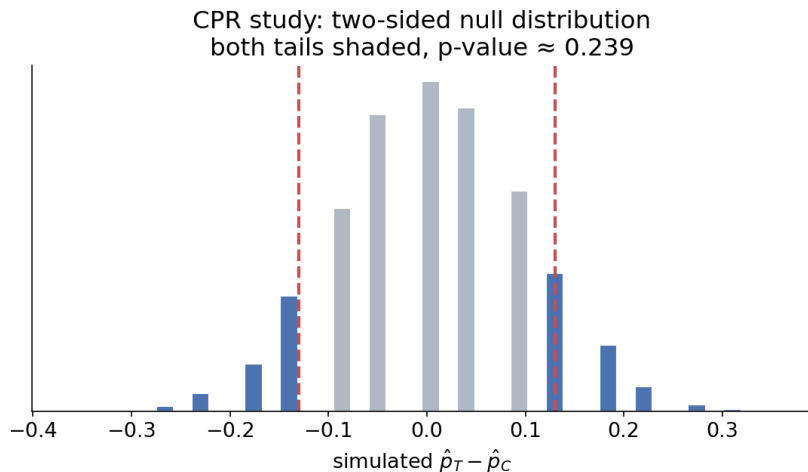
Case study: CPR and blood thinners

Does a blood thinner affect survival after CPR? $H_0 : p_T - p_C = 0$
vs. $H_A : p_T - p_C \neq 0$ (two-sided, since a thinner could plausibly help *or* hurt).

	Died	Survived	Total
Control	39	11	50
Treatment	26	14	40

$\hat{p}_C = 11/50 = 0.22$, $\hat{p}_T = 14/40 = 0.35$, observed difference = 0.13.

Computing a two-sided p-value



A difference of -0.13 would be *equally* strong evidence for H_A as $+0.13$ — so both tails count.

Never switch sides after seeing the data

Using $\alpha = 0.05$: what happens if we always pick whichever one-sided test matches the direction the data happen to point?

If the true difference is 0, about 5% of the time the sample difference lands high enough to reject with $H_A : \text{diff} > 0$, and *separately* about 5% of the time it lands low enough to reject with $H_A : \text{diff} < 0$.

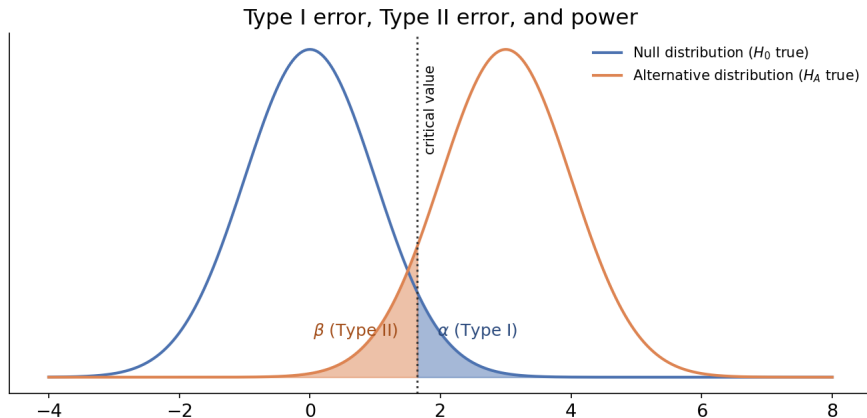
$$\text{True Type I error rate} = 5\% + 5\% = 10\% \neq 5\%$$

Hypotheses must be set up *before* looking at the data — otherwise your real error rate silently doubles.

Power

i Power

The probability of correctly rejecting H_0 when H_A is actually true.



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A well-designed study picks a sample size *in advance* large enough to have good power to detect an effect size that would actually matter.

Closing example: a real COVID-19 test

A 2024 study (Madewell et al., *Influenza and Other Respiratory Viruses*) evaluated the Abbott BinaxNOW rapid antigen card against RT-PCR in Puerto Rico:

*If you took this test today and it came back **positive**, what's the probability you actually have COVID? What if it came back **negative**?*

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The catch: the answer depends on something the test itself can't tell you — how common the disease is **right now**, in your community ($P(\text{disease})$). Let's see why that matters.

Same test, three prevalence scenarios

Using Bayes' theorem with $Se = 0.841$, $Sp = 0.988$, at three different community prevalence levels: (low) 2%, (moderate) 5%, and (outbreak) 20%, what would be the probability you have the disease given your tested positive (and negative)?

$$P(\text{disease} \mid +) = \frac{P(+ \mid \text{disease}) P(\text{disease})}{P(+ \mid \text{disease}) P(\text{disease}) + P(+ \mid \text{no disease}) P(\text{no disease})}$$

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Answers

Community prevalence	$P(\text{disease} \mid +)$	$P(\text{disease} \mid -)$
2% (low transmission)	58.9%	0.33%
5% (moderate)	78.7%	0.84%
20% (active outbreak)	94.6%	3.87%

- ▶ A **positive** result means something very different depending on how much virus is circulating — from “better than a coin flip either way” at low prevalence to “almost certainly infected” during an outbreak.
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The takeaway: the same test, with fixed sensitivity and specificity, gives genuinely different real-world information depending on context — which is exactly why public health guidance around testing changes over the course of an outbreak, even when the test itself hasn't changed at all.

Wrapping up

- ▶ Mathematical models (Ch. 13) give the *same* conclusions as simulation (Ch. 11-12) when conditions hold — and are much faster to compute — but they can fail silently when conditions (independence, large-enough n) aren't checked.
- ▶ Every hypothesis test risks two kinds of mistakes (Type I, Type II) — and reducing one generally increases the other.
- ▶ Default to **two-sided** tests unless you have a genuine, pre-specified reason not to.
- ▶ **Power** — our ability to detect a real effect — depends on effect size, sample size, and α , and is worth thinking about *before* collecting data, not after.